higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

FLUID MECHANICS N5

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This marking guideline consists of 10 pages.

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QUESTION 1

1.1 A real fluid is a fluid with viscosity.✓ The specific weight is the ratio of the weight per unit volume of a substance measured✓ in N/m^3.✓ (3)

1.2

$$\text{from}, \tau = \frac{\mu \Delta v}{t}$$

$$N_R = W_{perpendicular} = mg\cos\theta$$

$$¿ 31 \times 9,81 \times \cos 35 ✓$$

$$¿ 249,1123\ N ✓$$

$$\tau = \frac{mg\cos\theta}{A}$$

$$¿ \frac{249,1123}{0,95^2} ✓$$

$$¿ 276,02474\ \text{Pa} ✓$$

$$\text{now}, \mu = \frac{\tau t}{\Delta v}$$

$$\mu = \frac{276,02474 \times 2 \times 10^{-3}}{0,5} ✓$$

$$¿ 1,1041\ \text{Ns/m}^2 ✓$$

$$v = \frac{\mu}{\rho}$$

$$\rho = \frac{1,1041}{849,308 \times 10^{-6}} ✓$$

$$¿ 1\ 300\ \text{kg/m}^3 ✓$$

$$SG = \rho_{rel} = \frac{1\ 300}{1\ 000} ✓$$

$$¿ 1,3 ✓$$

(10)

1.3

$$\text{from}, v=\frac{\pi dN}{60}$$

$$t=\frac{D-d}{2}$$

$$=\frac{401,5-400}{2}$$

$$=0,75 \text{ mm} \checkmark$$

$$D_{mean}=\frac{D+d}{2}$$

$$¿\frac{401,5+400}{2}$$

$$¿400,75 \text{ mm} \checkmark$$

$$F=\frac{\mu Av}{t}$$

$$A=\pi DL$$

$$¿\pi\times0,40075\times0,095$$

$$¿0,1196 \ m^2 \checkmark$$

$$v=\frac{181\times0,75\times10^{-3}}{0,1196\times1,1041}\checkmark$$

$$¿1,02798 \text{ m/s} \checkmark$$

$$N=\frac{60\times1,02798}{\pi\times0,4}\checkmark$$

$$¿49,082 \text{ r/min} \checkmark$$

(7)
[20]

QUESTION 2

2.1 The coefficient of dynamic (or absolute) viscosity✓ is defined as the shear force per unit area required to drag one layer of fluid with unit velocity past another layer unit distance away from it in the fluid. The SI unit is Pa.s.✓ (2)

2.2 2.2.1 When a fluid is flowing in a system, the quantity of fluid entering✓ the system is the same as the quantity of fluid exiting✓ the system.

2.2.2 In a fluid-flow system, mass flow is the quantity of fluid measured✓ when flowing through or past an energy point in each unit of time.✓
(2 × 2) (4)

2.3 2.3.1

$$F=2\pi r\sigma$$

$$=2\pi\times0,01\times0,07$$

$$=439,823\times10^{-6}N \checkmark\checkmark$$

2.3.2

$$\Delta P = \frac{2\sigma}{r}$$
$$= \frac{2 \times 0,07}{0,001} \checkmark$$
$$= 140 \text{ Pa} \checkmark$$

(2 × 2) (4)

2.4 2.4.1

$$P = \rho g h$$
$$h = \frac{P}{\rho g} \checkmark$$
$$= \frac{150 \times 10^3}{9\,810}$$
$$\therefore 15,291 \ m \checkmark$$

2.4.2

$$h = \frac{P}{\rho g}$$
$$= \frac{150 \times 10^3}{1\,000 \times 0,83 \times 9,81}$$
$$\therefore 18,422 \ m \checkmark \checkmark$$

(2 × 2) (4)

2.5 2.5.1

$$P = \frac{\Delta F}{\Delta A}$$
$$\Delta A = \frac{\pi}{4}(0,08^2) - \left[\frac{\pi}{4}(0,08^2 - 0,02^2)\right] \checkmark$$
$$\therefore 0,000314 \ m^2 \checkmark$$
$$\Delta F = 150 \times 10^3 \times 0,000314 \checkmark$$
$$\therefore 47,124 \text{ kN} \checkmark$$

(4)

2.5.2

$$\text{from}, Q = \frac{V}{t}$$
$$t = \frac{\frac{\pi}{4}(0,08^2)}{200 \times 10^{-3}} \times 0,35 \checkmark$$
$$\therefore 8,796 \text{ sec} \checkmark$$

(2)
[20]

QUESTION 3

3.1 When a body is immersed in a liquid, it displaces a certain volume of the liquid.✓ The liquid induces an upward thrust (force) equal to the weight of the fluid displaced by the floating body.✓ The upward thrust✓ balances the weight of the floating body. The higher the density of the liquid the greater the upward thrust✓ for the same quantity of volume displaced.✓ (5)

3.2 3.2.1

$$W_{object} = F_{bouyancy}$$
$$W = \rho g V \checkmark$$
$$\therefore 9\,810 \times \frac{\pi \times 1{,}2^2}{4} \times 3 \times 0{,}7 \checkmark$$
$$\therefore 23\,299{,}182\ N \checkmark$$ (3)

3.2.2

$$W_{object} = F_{bouyancy}$$
$$W = \rho g V_p$$
$$\therefore 9\,810 \times \frac{\pi \times 1{,}2^2}{4} \times 3 \checkmark$$
$$\therefore 33\,284{,}546\ N \checkmark$$

$$V_p = \frac{33\,284{,}546}{7\,850 \times 9{,}81} \checkmark$$
$$\therefore 0{,}432\ m^3 \checkmark$$

$$\text{also}, V_p = \pi DLt + 2\frac{\pi}{4} D^2 t$$
$$0{,}432 = \pi \times 1{,}2(3 - 2t)t + \left(2 \times \frac{\pi}{4} \times 1{,}2^2 t\right) \checkmark$$
$$0{,}432 = 11{,}3097t + 2{,}262t - 7{,}5398t^2 \checkmark$$
$$\Rightarrow 7{,}5398t^2 - 13{,}572t - 0{,}432 = 0 \checkmark$$
$$\therefore t = 0{,}0324\ m \checkmark$$ (8)

3.2.3

$$F = \rho g \bar{y} A$$
$$\bar{y} = \frac{1{,}2}{2} \checkmark$$
$$\therefore 0{,}6\ m \checkmark$$
$$\therefore F = 9\,810 \times 0{,}6 \times \frac{\pi}{4} \times 1{,}2^2 \checkmark$$
$$\therefore 6\,656{,}909\ N \checkmark$$ (4)

[20]

QUESTION 4

4.1 The flow field region where the velocity flow is in disorder or not uniform in nature is the turbulent region due to its low resistance to motion due to a low✓ (negligible) viscosity,✓ e.g. a gas. While in the lamina region the fluid flows in an orderly manner due to the high viscosity of the fluid,✓ e.g. grease.✓ (4)

4.2 $\text{from , } R_e = \frac{\rho v d}{\mu}$

$\upsilon = \frac{\mu}{\rho}$

$\mu = 0,82 \times 1\ 000 \times 24,39 \times 10^{-6}$ ✓

$R_e = 0,0199998\ \text{Pa}.s$ ¿

¿ $\frac{820 \times 0,1 \times 0,1}{0,0199998}$ ✓

$\therefore R_e = 410,004$

because R_e ¿✓

¿ ✓ (5)

4.3 4.3.1 from Bernoulli's equation

$E_1 - H_{loss} = E_2$

$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + Z_1 - H_{loss} = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + Z_2$ ✓

$\frac{100 \times 10^3}{820 \times 9,81} - 5 = \frac{v_2^2}{2g} + 3$ ✓ ✓

$v = \sqrt{(12,43137 - 8) \times 19,62}$ ✓

$v = 9,324\ \text{m/s}$ ✓

$Q = Av$

¿ $\frac{\pi \times 0,1^2}{4} \times 9,324$ ✓

¿ $0,0732\ m^3/\text{s}$ ✓ (7)

4.3.2 $Q = \frac{V}{t}$

$t = \frac{200}{0,0732}$ ✓

$= 2\ 731,016\ \text{sec}$ ✓

$= 45,517\ \text{min}$ ✓ (3)

4.3.3 ^{0}

$m = \rho Q$

¿ $820 \times 0,225$

¿ $184,5 \text{ kg/min}$ ✓ (1)

[20]

QUESTION 5

5.1 A venturi flowmeter is a mechanical device used to measure✓ velocity of flow in a fluid pipeline system due to the pressure changes✓ (difference) between the entrance and throat sections of the meter.✓ (3)

5.2

$\text{from}, PE = mgh \text{ and } EK = \frac{mv^2}{2}$

$\text{since}, PE = EK$

$\therefore h = \frac{v^2}{2g}$ ✓

$h = \frac{13,889^2}{19,62}$ ✓

¿ $9,832 \text{ m}$ ✓

$P_{tube} + P_{meter} = \rho gh$

$(9\,810 \times 9,832) = P_{meter} + (9\,810 \times 1)$ ✓✓

$\therefore P_{meter} = 86,641 \text{ kPa}$ ✓ (6)

5.3 5.3.1 consider all the components

consider 90° bend:

$$l/d=\left(\frac{k}{4f}\right)$$

$$¿2\left(\frac{0,75}{4\times0,002}\right)\checkmark$$

$$¿187,5\checkmark$$

consider filter:

$$l/d=\left(\frac{3}{4\times0,002}\right)$$

$$¿375\ \checkmark$$

regulating valve:

$$l/d=\left(\frac{0,64}{4\times0,002}\right)$$

$$¿80$$

pipe: ✓

$$l/d=\frac{10}{0,025}$$

$$¿400\ \checkmark$$

$$l/d\,(system)=400+80+375+187,5\ \checkmark$$

$$¿1\ 024,5\checkmark$$

(7)

5.3.2

$$hf\,(system)=\frac{4\,flv^2}{2\,gd}$$

$$v=\frac{4\times1,2\times10^{-3}}{\pi\times0,025^2}\ \checkmark$$

$$¿2,44462\ \text{m/s}\ \checkmark$$

$$\therefore hf\,(system)=\frac{4\times0,002\times2,44462^2}{19,62}\times1\ 024,5\ \checkmark$$

$$¿2,495\ m\ \checkmark$$

(4)

[20]

QUESTION 6

6.1

$$\text{from}, \overset{0}{W} = \overset{0}{m}g$$
$$\text{where}, Q = Av$$
$$¿ \frac{\pi}{4} \times 0,15^2 \times 2$$
$$¿ 35,343 \times 10^{-3} m^3/\text{s} \checkmark$$
$$\text{also}, \overset{0}{m} = \rho Q$$
$$¿ 1\ 000 \times 35,343 \times 10^{-3}$$
$$¿ 35,343 \text{ kg/s} \checkmark$$
$$\therefore \overset{0}{W} = 35,343 \times 9,81$$
$$¿ 346,714 \text{ N/s} \checkmark$$

(3)

6.2

$$\text{from}, \overset{0}{W} = \overset{0}{m}g$$
$$\text{where}, Q = Av$$
$$v_2 = \frac{35,343 \times 10^{-3}}{\frac{\pi}{4} \times 0,15^2} \checkmark$$
$$¿ 4,5\ m^3/\text{s} \checkmark$$

(2)

6.3

$$E_{inlet} - hf = E_{exit}$$
$$\frac{P_{inlet}}{\rho g} + \frac{v_{inlet}^2}{2g} + Z_{inlet} = \frac{P_{exit}}{\rho g} + \frac{v_{exit}^2}{2g} + Z_{exit} \checkmark$$
$$\frac{300 \times 10^3}{9\ 810} + \frac{2^2}{19,62} + 0 - 2 = \frac{P_{exit}}{\rho g} + \frac{4,5^2}{19,62} + 3 \checkmark$$
$$P_{exit} = 24,7528 \times 9\ 810 \checkmark$$
$$\therefore P_{exit} = 242,825 \text{ kPa} \checkmark$$

(4)

6.4

$$P_{loss} = \overset{0}{W} h_{loss}$$
$$¿ 346,7 \times 2 \checkmark$$
$$¿ 693,428\ W \checkmark$$

(2)

6.5

$$F_{inlet} = \overset{0}{m}_{inlet} v_{inlet} + P_{inlet} A_{inlet}$$

$$¿ (35,343 \times 2) + \left(300 \times 10^3 \times \frac{\pi}{4} \times 0,15^2\right) \checkmark$$

$$¿ 5,372 \text{ kN} \checkmark$$

$$F_{exit} = \overset{0}{m}_{exit} v_{exit} + P_{exit} A_{exit}$$

$$¿ (35,343 \times 4,5) + \left(242,825 \times 10^3 \times \frac{\pi}{4} \times 0,1^2\right) \checkmark$$

$$¿ 2,066 \text{ kN} \checkmark$$ (4)

6.6

$$F_{res} = \sqrt{F_x^2 + F_y^2}$$

$$¿ \sqrt{(5,372 - 2,066\cos 55)^2 + (2,066\sin 55)^2} \checkmark\checkmark$$

$$¿ 4,516 \text{ kN} \checkmark$$

$$\tan \beta = \frac{F_y}{F_x}$$

$$\beta = \tan^{-1}\left(\frac{2,066\sin 55}{5,372 - 2,066\cos 55}\right) \checkmark$$

$$¿ 22,008^\circ \checkmark$$ (5)

[20]

TOTAL: 100

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